

USEPA DMR-QA 33

NPDES PERMITTEE DATA REPORT FORM



ERA

RECEIVED
A. Waters Company

USEPA NPDES MI0020257

Permit #:

Permit Ext:

AUG 14 2013

Permittee Name: Cadillac WWTP

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Facility Address: 1121 Plett Rd.

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

Fax Number: (231) 775-2876

E-mail address: lab@cadillac-mi.net

For DMR-QA Study 33, conducted in 2013, the Permittee ensured that their laboratory(s) performing the required analyses:

Received PT Samples

Yes ☒ No ☐

Submitted Complete and Accurate Data
by July 8, 2013

Yes ☒ No ☐

Received a Graded Report by July
29, 2013

Yes ☒ No ☐

Certification by Permit Holder or Authorized Representative
(as per 40 C.F.R. Section 122.22)

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate and complete. Each reported value was produced from a single analytical run using the analytical system that routinely performs these analyses to produce compliance monitoring data required under our National Pollutant Discharge Elimination System (NPDES) Permit. Neither I nor any of my subordinates compared our results from independent analyses conducted by us or any other laboratory before we reported our results to the USEPA. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name of
Certifying Official: Amy Vall

Title: Lab Manager

Signature:

Date Signed:

8/19/2013

Mailing Address: 200 N Lake Street

(enter only if different
from address above)

City: Cadillac

State: MI

Postal Code: 49601

Phone Number: (231) 775-2368

E-mail address: lab@cadillac-mi.net

Please make corrections in boxes above

United States

ENVIRONMENTAL PROTECTION AGENCYLaboratory Performance Evaluation
Laboratory DMR-QA Evaluation Study 33

USEPA NPDES Permit #: MI0020257

Permit Ext:

Identification of all CHEM, MICRO and TOX laboratories who did analyses for this permit

Name of Laboratory	Address of Laboratory	U.S. EPA Lab Code	Lab Analysis Check box(es) that apply			Lab Type*	State-certified Lab**
			CHEM	MICRO	TOX		
Cadillac WWTP	200 N. Lake St. Cadillac, MI 49601	MI00082	☒	☒	☐	G	☒
Kar Labs	4425 Manchester Rd. Kalamazoo, MI 49001	MI00041	☒	☐	☐	C	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒
Paragon Laboratories, Inc.	12649 Richfield Ct Livonia, MI 48150	MI01187	☐	☐	☒	C	☒

* Lab Types: C = Commercial F = Federal G = Local Government I = Industrial O = Other S = State

** See Footnote 2 on DMRQA-33, Frequently Asked Questions page

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist**DMRQA Study 33**

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Chloride Alkalinity as CaCO ₃ Sodium Total Dissolved Solids at 180°C Conductivity at 25°C Potassium Total Solids at 105°C Fluoride Sulfate				
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃				
pH pH				
Settleable Solids Settleable Solids				
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Ion-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Nitrate + Nitrite as N Nitrate as N ortho-Phosphate as P Ammonia as N				
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P				
Nitrite Nitrite as N				
Demand COD TOC CBOD BOD				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
<u>Oil & Grease Concentrate</u>				
n-Hexane Extractable Material(O&G) (Grav)				
<u>Oil & Grease</u>				
n-Hexane Extractable Material(O&G) (IR)				
n-Hexane Extractable Material(O&G) (Grav)				
<u>Trace Metals</u>				
Antimony				
Cadmium				
Silver				
Lead				
Molybdenum				
Selenium				
Thallium				
Barium				
Copper				
Iron				
Nickel				
Strontium				
Arsenic				
Cobalt				
Zinc				
Aluminum				
Beryllium				
Chromium				
Vanadium				
Boron				
Manganese				
<u>Mercury</u>				
Mercury				
<u>Low-Level Mercury</u>				
Low Level Mercury	☐	☒	☐	☐
<u>Hexavalent Chromium</u>				
Hexavalent Chromium				
<u>Turbidity</u>				
Turbidity				
<u>Total Cyanide</u>				
Cyanide, total				
<u>Total Phenolics (4-AAP)</u>				
Phenolics, total				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00041

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name,

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Chloride Alkalinity as CaCO ₃ Sodium Total Dissolved Solids at 180°C Conductivity at 25°C Potassium Total Solids at 105°C Fluoride Sulfate				
Hardness Non-Filterable Residue (TSS) SM 2540 D-2011 Calcium Magnesium Calcium Hardness as CaCO ₃ Total Hardness as CaCO ₃	☐	☐	☒	☐
pH pH SM 4500-H+ B-2011	☐	☒	☐	☐
Settleable Solids Settleable Solids				
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Nitrate as N Nitrate + Nitrite as N Ammonia as N SM 4500-NO ₃ D-2011 ortho-Phosphate as P	☐	☒	☐	☐
Complex Nutrients Total phosphorus as P SM 4500-P E-2011 Total Kjeldahl Nitrogen	☐	☒	☐	☐
Nitrite Nitrite as N				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Demand BOD CBOD SM 5210 B-2011 COD TOC	☐	☒	☐	☐
Oil & Grease Concentrate n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease n-Hexane Extractable Material(O&G) (IR) n-Hexane Extractable Material(O&G) (Grav)				
Trace Metals Barium Cadmium Iron Silver Antimony Lead Molybdenum Selenium Thallium Copper Nickel Zinc Cobalt Arsenic Vanadium Beryllium Aluminum Chromium Strontium Boron Manganese				
Mercury Mercury				
Low-Level Mercury Low Level Mercury				
Hexavalent Chromium Hexavalent Chromium				
Turbidity Turbidity				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI00082

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Cyanide				
Cyanide, total				
Total Phenolics (4-AAP)				
Phenolics, total				
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
SM 9222 D (m-FC)-1997	☐	☒	☐	☐
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals Potassium Fluoride Sulfate Total Solids at 105°C Chloride Alkalinity as CaCO3 Sodium Total Dissolved Solids at 180°C Conductivity at 25°C				
Hardness Non-Filterable Residue (TSS) Calcium Magnesium Calcium Hardness as CaCO3 Total Hardness as CaCO3 EPA 130.1	☐	☒	☐	☐
pH pH				
Settleable Solids Settleable Solids				
Solids Concentrate Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Solids Non-Filterable Residue (TSS) Total Dissolved Solids at 180°C Total Solids at 105°C				
Simple Nutrients Nitrate as N ortho-Phosphate as P Nitrate + Nitrite as N Ammonia as N				
Complex Nutrients Total Kjeldahl Nitrogen Total phosphorus as P				
Nitrite Nitrite as N				
Demand COD TOC CBOD BOD				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Oil & Grease Concentrate				
n-Hexane Extractable Material(O&G) (Grav)				
Oil & Grease				
n-Hexane Extractable Material(O&G) (Grav)				
n-Hexane Extractable Material(O&G) (IR)				
Trace Metals				
Chromium				
Vanadium				
Boron				
Manganese				
Aluminum				
Beryllium				
Cadmium				
Silver				
Lead				
Molybdenum				
Selenium				
Thallium				
Antimony				
Barium				
Copper				
Iron				
Nickel				
Strontium				
Arsenic				
Cobalt				
Zinc				
Mercury				
Mercury				
Low-Level Mercury				
Low Level Mercury				
Hexavalent Chromium				
Hexavalent Chromium				
Turbidity				
Turbidity				
Total Cyanide				
Cyanide, total				
Total Phenolics (4-AAP)				
Phenolics, total				
Total Residual Chlorine				
Total Residual Chlorine				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

DMRQA Study 33

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMR-QA Study 33

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
Fathead minnow (Test Code 13)						
0754	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2000.0	LC50	☐	☒	☐	☐
Fathead minnow (Test Code 14)						
0755	48Hr., Acute, Non-Renewal, 25° C, 20% DMW	LC50				
Fathead minnow (Test Code 15)						
0808	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000.0	IC25 (ON) Growth	☐	☒	☐	☐
0810	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000.0	NOEC (ON) Growth	☐	☒	☐	☐
0756	7-day Short term Chronic, Daily Renewal, MHSF EPA 1000.0	NOEC Survival	☐	☒	☐	☐
Fathead minnow (Test Code 16)						
0812	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 (ON) Growth				
0814	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC (ON) Growth				
0759	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
riodaphnia dubia (Test Code 19)						
0764	48Hr., Acute Renewal, 25° C, MHSF EPA 2002.0	LC50	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 20)						
0765	48Hr., Acute Renewal, 25° C, 20% DMW	LC50				
Ceriodaphnia dubia (Test Code 21)						
0767	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002.0	IC25 Reproduction	☐	☒	☐	☐
0768	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002.0	NOEC Reproduction	☐	☒	☐	☐
0766	7-day Short term Chronic, Daily Renewal, MHSF EPA 1002.0	NOEC Survival	☐	☒	☐	☐
Ceriodaphnia dubia (Test Code 22)						
0770	7-day Short term Chronic, Daily Renewal, 20% DMW	IC25 Reproduction				
0771	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Reproduction				
0769	7-day Short term Chronic, Daily Renewal, 20% DMW	NOEC Survival				
Daphnia magna (Test Code 32)						
0788	48Hr., Acute, Non-Renewal, 25° C, MHSF EPA 2021.0	LC50	☐	☒	☐	☐

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

WET Organisms/Test Conditions/End Points Checklist

DMR-QA Study 33

Analyte Number	Organisms / Conditions	End Points	Test Req	Laboratory's Graded Result		Analyte determined by state-certified lab
				Acceptable	Not Acceptable (Corrective Action Required)	
<u>Daphnia pulex (Test Code 38)</u>						
0794	48Hr., Acute, Non-Renewal, 25° C, MHSF	LC50				
<u>Mysid (Test Code 42)</u>						
0798	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Mysid (Test Code 43)</u>						
0816	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0818	7-day Short term Chronic, Daily Renewal, 40 FSW					
0799	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Inland silverside (Test Code 44)</u>						
0803	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Inland Silverside (Test Code 45)</u>						
0825	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0826	7-day Short term Chronic, Daily Renewal, 40 FSW					
0824	7-day Short term Chronic, Daily Renewal, 40 FSW					
<u>Sheepshead minnow (Test Code 46)</u>						
0804	48Hr., Acute, Non-Renewal, 25° C, 40 FSW	LC50				
<u>Sheepshead minnow (Test Code 47)</u>						
0820	7-day Short term Chronic, Daily Renewal, 40 FSW	IC25 (ON) Growth NOEC (ON) Growth NOEC Survival				
0822	7-day Short term Chronic, Daily Renewal, 40 FSW					
0805	7-day Short term Chronic, Daily Renewal, 40 FSW					

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 219

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals				
Fluoride EPA 300.0	☐	☒	☐	☐
Potassium EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Sodium EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Sulfate EPA 300.0	☐	☒	☐	☐
Total Dissolved Solids at 180°C SM 2540 C-1997	☐	☒	☐	☐
Alkalinity as CaCO3 SM 2320 B-1997	☐	☒	☐	☐
Conductivity at 25°C SM 2510 B-1997	☐	☒	☐	☐
Total Solids at 105°C Chloride EPA 300.0	☐	☒	☐	☐
Hardness				
Non-Filterable Residue (TSS)				
Calcium EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Magnesium EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Calcium Hardness as CaCO3 EPA 200.8	☐	☒	☐	☐
SM 2340 B-1997	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Total Hardness as CaCO3 SM 2340 C-1997	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
SM 2340 B-1997	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
pH				
pH SM 4500-H+ B-2000	☐	☒	☐	☐

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 219

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Settleable Solids				
Settleable Solids				
Solids Concentrate				
Non-Filterable Residue (TSS)				
Total Dissolved Solids at 180°C				
Total Solids at 105°C				
Solids				
Non-Filterable Residue (TSS)				
Total Dissolved Solids at 180°C				
Total Solids at 105°C				
Simple Nutrients				
Nitrate + Nitrite as N				
EPA 300.0	☐	☒	☐	☐
EPA 353.2	☐	☒	☐	☐
ortho-Phosphate as P				
EPA 365.3	☐	☒	☐	☐
Ammonia as N				
SM 4500-NH3 G-2011	☐	☒	☐	☐
Nitrate as N				
EPA 353.2	☐	☒	☐	☐
EPA 300.0	☐	☒	☐	☐
Complex Nutrients				
Total phosphorus as P				
EPA 365.3	☐	☒	☐	☐
Total Kjeldahl Nitrogen				
Nitrite				
Nitrite as N				
EPA 300.0	☐	☒	☐	☐
EPA 353.2	☐	☒	☐	☐
Demand				
BOD				
SM 5210 B-2001	☐	☒	☐	☐
COD				
EPA 410.4	☐	☒	☐	☐
CBOD				
SM 5210 B-2001	☐	☒	☐	☐
TOC				
SM 5310 C-2000	☐	☒	☐	☐
Oil & Grease Concentrate				
n-Hexane Extractable Material(O&G) (Grav)				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 219

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Oil & Grease				
n-Hexane Extractable Material(O&G) (Grav)	☐	☒	☐	☐
EPA 1664A				
n-Hexane Extractable Material(O&G) (IR)				
Trace Metals				
Barium				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Cadmium				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Copper				
EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Iron				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Silver				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Strontium				
EPA 200.8	☐	☒	☐	☐
Thallium				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Vanadium				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Boron				
EPA 200.8	☐	☒	☐	☐
Chromium				
EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Cobalt				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Nickel				
EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Selenium				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 219

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
EPA 200.8	☐	☒	☐	☐
Arsenic				
EPA 200.8	☐	☒	☐	☐
Beryllium				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Molybdenum				
EPA 200.7	☐	☒	☐	☐
EPA 200.8	☐	☒	☐	☐
Zinc				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Antimony				
EPA 200.8	☐	☒	☐	☐
Manganese				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Aluminum				
EPA 200.8	☐	☒	☐	☐
EPA 200.7	☐	☒	☐	☐
Lead				
EPA 200.8	☐	☒	☐	☐
Mercury				
Mercury				
EPA 245.1	☐	☒	☐	☐
Low-Level Mercury				
Low Level Mercury				
EPA 1631E	☐	☒	☐	☐
Hexavalent Chromium				
Hexavalent Chromium				
SM 3500-Cr B-2009	☐	☒	☐	☐
Turbidity				
Turbidity				
SM 2130 B-2001	☐	☒	☐	☐
EPA 180.1	☐	☒	☐	☐
Total Cyanide				
Cyanide, total				
EPA 335.4	☐	☒	☐	☐
Amenable Cyanide				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 219

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Phenolics (4-AAP)				
Phenolics, total EPA 420.4	☐	☒	☐	☐
Total Residual Chlorine				
Total Residual Chlorine SM 4500-Cl G-2000	☐	☒	☐	☐
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF) EPA 1604 MIAgar	☐	☒	☐	☐
Fecal Coliforms (MF) E.coli (MF) EPA 1604 MIAgar	☐	☒	☐	☐
Total Coliforms (MPN) Fecal Coliforms (MPN) E.coli (MPN)				

Print Name

Signature/Title

Date

Use a separate checklist for EACH lab used

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 220

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Minerals				
Alkalinity as CaCO ₃ EPA 310.2	☐	☒	☐	☐
Fluoride				
Sulfate				
Total Dissolved Solids at 180°C				
Sodium				
Chloride				
Conductivity at 25°C				
Potassium				
Total Solids at 105°C SM 2540 B-1997	☐	☒	☐	☐
Hardness				
Non-Filterable Residue (TSS) SM 2540 D-1997	☐	☒	☐	☐
Calcium				
Magnesium				
Calcium Hardness as CaCO ₃				
Total Hardness as CaCO ₃				
pH				
pH				
Settleable Solids				
Settleable Solids				
Solids Concentrate				
Non-Filterable Residue (TSS)				
Total Dissolved Solids at 180°C				
Total Solids at 105°C				
Solids				
Non-Filterable Residue (TSS)				
Total Dissolved Solids at 180°C				
Total Solids at 105°C				
Simple Nutrients				
Nitrate as N				
Nitrate + Nitrite as N				
Ammonia as N				
ortho-Phosphate as P				
Complex Nutrients				
Total Kjeldahl Nitrogen				
Total phosphorus as P				
Nitrite				
Nitrite as N				
Demand				
BOD				

Permittee Name: Cadillac WWTP

Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 220

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
CBOD COD TOC				
<u>Oil & Grease Concentrate</u> n-Hexane Extractable Material(O&G) (Grav)				
<u>Oil & Grease</u> n-Hexane Extractable Material(O&G) (Grav) n-Hexane Extractable Material(O&G) (IR)				
<u>Trace Metals</u> Arsenic Vanadium Barium Cadmium Iron Antimony Boron Lead Copper Nickel Zinc Cobalt Aluminum Silver Strontium Thallium Chromium Manganese Molybdenum Selenium Beryllium				
<u>Mercury</u> Mercury				
<u>Hexavalent Chromium</u> Hexavalent Chromium				
<u>Turbidity</u> Turbidity				
<u>Total Cyanide</u> Cyanide, total Amenable Cyanide				
<u>Total Phenolics (4-AAP)</u> Phenolics, total				

Permittee Name: Cadillac WWTP

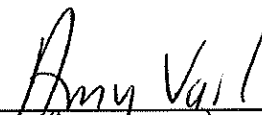
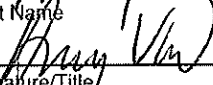
Permit Number: MI0020257

EPA Lab Code: MI01187

Chemistry/Microbiology Analyte Checklist

WP Study 220

Analyte Test / Method	Test Required	Laboratory's Graded Result		Analyte determined by state-certified lab
		Acceptable	Not Acceptable (Corrective Action Required)	
Total Residual Chlorine				
Total Residual Chlorine				
Low-Level Total Residual Chlorine				
Low Level Total Residual Chlorine				
WasteWatR™ Coliform MicrobE™				
Total Coliforms (MF)				
Fecal Coliforms (MF)				
E.coli (MF)				
Total Coliforms (MPN)				
Fecal Coliforms (MPN)				
E.coli (MPN)				


 Print Name
 Lab Manager
 Signature/Title

8/9/13
 Date

Use a separate checklist for EACH lab used



A Waters Company

Study: **DMR-QA 33**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

Inorganic Results





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: M10020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Minerals (cat# 581)											
1505	Alkalinity as CaCO ₃	mg/L	Acceptable	37.0	37.7	32.0 - 44.2	SM 2320 B-1997 1997	37.3	1.90	M101187	WP219
1575	Chloride	mg/L	Acceptable	111.	116	99.8 - 132	EPA 300.0 2.1 1993	113	4.18	M101187	WP219
1610	Conductivity at 25 °C	µmhos/cm	Acceptable	476	515	464 - 567	SM 2510 B-1997 1997	515	10.4	M101187	WP219
1730	Fluoride	mg/L	Acceptable	1.24	1.29	1.01 - 1.57	EPA 300.0 2.1 1993	1.22	0.0987	M101187	WP219
1125	Potassium	mg/L	Acceptable	19.5	20.8	17.1 - 24.8	EPA 200.7 4.4 1994	20.1	1.23	M101187	WP219
1155	Sodium	mg/L	Acceptable	93.4	92.0	78.1 - 105	EPA 200.7 4.4 1994	88.8	5.24	M101187	WP219
2000	Sulfate	mg/L	Acceptable	20.6	22.2	17.4 - 26.4	EPA 300.0 2.1 1993	21.2	1.10	M101187	WP219
1955	Total Dissolved Solids at 180 °C	mg/L	Acceptable	294.	301	226 - 376	SM 2540 C-1997 1997	301	13.8	M101187	WP219
WP Minerals (cat# 581)											
1125	Potassium	mg/L	Acceptable	19.5	20.8	17.1 - 24.8	EPA 200.8 5.4 1994	20.1	1.23	M101187	WP219
1155	Sodium	mg/L	Acceptable	86.1	92.0	78.1 - 105	EPA 200.8 5.4 1994	88.8	5.24	M101187	WP219
WP Minerals (cat# 581)											
1505	Alkalinity as CaCO ₃	mg/L	Acceptable	50.7	53.0	46.0 - 60.2	EPA 310.2 1974	52.3	2.11	M101187	WP220
1950	Total Solids at 105 °C	mg/L	Acceptable	244.	235	198 - 270	SM 2540 B-1997 1997	232	15.8	M101187	WP220
DMRQA Hardness (cat# 580)											
1960	Non-Filterable Residue (TSS)	mg/L	Not Acceptable	67.3	56.6	44.6 - 64.2	SM 2540 D-2011 2011	52.0	3.77	M100082	DMRQA33
WP Hardness (cat# 580)											
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	100.	109	95.0 - 124	SM 2340 C-1997 1997	106	4.50	M101187	WP219
WP Hardness (cat# 580)											
1550	Calcium Hardness as CaCO ₃	mg/L	Acceptable	70.6	74.5	66.1 - 84.9	SM 2340 B-1997 1997	72.9	2.96	M101187	WP219
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	104.	109	95.0 - 124	SM 2340 B-1997 1997	106	4.50	M101187	WP219

All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01



Page 2 of 12

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Hardness (cat# 580)											
1550	Calcium Hardness as CaCO ₃	mg/L	Acceptable	72.4	74.5	66.1 - 84.9	SM 2340 B-1997 (MS)	72.9	2.96	MI01187	WP219
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	107.	109	95.0 - 124	SM 2340 B-1997 (MS)	106	4.50	MI01187	WP219
WP Hardness (cat# 580)											
1035	Calcium	mg/L	Acceptable	28.3	29.8	26.4 - 34.0	EPA 200.7 4.4 1994	29.1	1.34	MI01187	WP219
1085	Magnesium	mg/L	Acceptable	8.02	8.30	7.04 - 9.50	EPA 200.7 4.4 1994	8.15	0.450	MI01187	WP219
1550	Calcium Hardness as CaCO ₃	mg/L	Acceptable	70.6	74.5	66.1 - 84.9	EPA 200.7 4.4 1994	72.9	2.96	MI01187	WP219
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	104.	109	95.0 - 124	EPA 200.7 4.4 1994	106	4.50	MI01187	WP219
WP Hardness (cat# 580)											
1035	Calcium	mg/L	Acceptable	27.5	29.8	26.4 - 34.0	EPA 200.8 5.4 1994	29.1	1.34	MI01187	WP219
1085	Magnesium	mg/L	Acceptable	7.95	8.30	7.04 - 9.50	EPA 200.8 5.4 1994	8.15	0.450	MI01187	WP219
1550	Calcium Hardness as CaCO ₃	mg/L	Acceptable	72.4	74.5	66.1 - 84.9	EPA 200.8 5.4 1994	72.9	2.96	MI01187	WP219
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	107.	109	95.0 - 124	EPA 200.8 5.4 1994	106	4.50	MI01187	WP219
WP Hardness (cat# 580)											
1960	Non-Filterable Residue (TSS)	mg/L	Acceptable	35.0	35.9	26.3 - 42.2	SM 2540 D-1997 1997	32.9	3.05	MI01187	WP220
DMRQA Hardness (cat# 580)											
1755	Total Hardness as CaCO ₃	mg/L	Acceptable	203.	217	189 - 247	EPA 130.1 1982	212	8.23	MI01187	DMRQA33
DMRQA pH (cat# 577)											
1900	pH	S.U.	Acceptable	5.96	5.90	5.70 - 6.10	SM 4500-H+ B-2011 2011	5.95	0.0642	MI00082	DMRQA33
WP pH (cat# 577)											
1900	pH	S.U.	Acceptable	7.2	7.14	6.94 - 7.34	SM 4500-H+ B-2000 2000	7.16	0.0570	MI01187	WP219



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Simple Nutrients (cat# 584)											
1515	Ammonia as N	mg/L	Acceptable	12.1	13.5	10.0 - 16.8	SM 4500-NH3 D-2011	13.4	1.03	MI00082	DMRQA33
WP Simple Nutrients (cat# 584)											
1515	Ammonia as N	mg/L	Acceptable	7.96	10.1	7.48 - 12.6	SM 4500-NH3 G-2011	10.0	0.617	MI01187	WP219
1820	Nitrate + Nitrite as N	mg/L	Acceptable	4.10	4.37	3.56 - 5.09	EPA 300.0 2.1 1993	4.31	0.186	MI01187	WP219
1810	Nitrate as N	mg/L	Acceptable	4.00	4.37	3.40 - 5.29	EPA 300.0 2.1 1993	4.30	0.200	MI01187	WP219
1870	ortho-Phosphate as P	mg/L	Acceptable	5.27	5.15	4.26 - 6.08	EPA 365.3 1978	5.10	0.233	MI01187	WP219
WP Simple Nutrients (cat# 584)											
1820	Nitrate + Nitrite as N	mg/L	Acceptable	4.55	4.37	3.56 - 5.09	EPA 353.2 2 1993	4.31	0.186	MI01187	WP219
1810	Nitrate as N	mg/L	Acceptable	4.55	4.37	3.40 - 5.29	EPA 353.2 2 1993	4.30	0.200	MI01187	WP219
DMRQA Complex Nutrients (cat# 579)											
1910	Total phosphorus as P	mg/L	Acceptable	3.45	3.31	2.69 - 3.98	SM 4500-P E-2011	3.37	0.237	MI00082	DMRQA33
WP Complex Nutrients (cat# 579)											
1910	Total phosphorus as P	mg/L	Acceptable	2.28	2.30	1.85 - 2.80	EPA 365.3 1978	2.35	0.144	MI01187	WP219
WP Nitrite (cat# 888)											
1840	Nitrite as N	mg/L	Acceptable	2.89	3.28	2.79 - 3.77	EPA 300.0 2.1 1993	3.16	0.181	MI01187	WP219
WP Nitrite (cat# 888)											
1840	Nitrite as N	mg/L	Acceptable	3.15	3.28	2.79 - 3.77	EPA 353.2 1993	3.16	0.181	MI01187	WP219
DMRQA Demand (cat# 578)											
1555	CBOD	mg/L	Acceptable	89.6	71.9	32.2 - 112	SM 5210 B-2011 2011	78.9	15.9	MI00082	DMRQA33

All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: M10020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Demand (cat# 578)											
1530	BOD	mg/L	Acceptable	29.1	38.3	19.1 - 57.5	SM 5210 B-2001 2001	37.8	6.79	MI01187	WP219
1555	CBOD	mg/L	Acceptable	38.4	33.0	14.8 - 51.3	SM 5210 B-2001 2001	35.0	6.83	MI01187	WP219
1565	COD	mg/L	Acceptable	64.0	61.7	43.1 - 75.3	EPA 410.4.2 1993	60.7	5.25	MI01187	WP219
2040	TOC	mg/L	Acceptable	23.1	24.4	20.3 - 28.4	SM 5310 C-2000 2000	24.4	1.13	MI01187	WP219
WP Oil & Grease (cat# 582)											
1660	n-Hexane Extractable Material(O&G)(Grav)	mg/L	Acceptable	82.7	90.0	63.2 - 105	EPA 1664A	84.0	8.42	MI01187	WP219



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



Page 5 of 12



DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: MI0020257
Permit Holder: Amy Vail
Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Trace Metals (cat# 586)											
1000	Aluminum	µg/L	Acceptable	332.	316	232 - 403	EPA 200.8.5.4 1994	331	22.8	MI01187	WP219
1005	Antimony	µg/L	Acceptable	377.	388	268 - 469	EPA 200.8.5.4 1994	378	21.0	MI01187	WP219
1010	Arsenic	µg/L	Acceptable	550.	564	473 - 660	EPA 200.8.5.4 1994	551	30.4	MI01187	WP219
1015	Barium	µg/L	Acceptable	1940.	1900	1550 - 2140	EPA 200.8.5.4 1994	1890	77.7	MI01187	WP219
1020	Beryllium	µg/L	Acceptable	43.1	40.3	32.9 - 45.7	EPA 200.8.5.4 1994	39.7	2.26	MI01187	WP219
1025	Boron	µg/L	Acceptable	1540.	1450	1180 - 1690	EPA 200.8.5.4 1994	1440	79.8	MI01187	WP219
1030	Cadmium	µg/L	Acceptable	584.	608	519 - 690	EPA 200.8.5.4 1994	597	24.5	MI01187	WP219
1040	Chromium	µg/L	Acceptable	340.	358	311 - 406	EPA 200.8.5.4 1994	358	15.6	MI01187	WP219
1050	Cobalt	µg/L	Acceptable	423.	424	372 - 476	EPA 200.8.5.4 1994	438	17.7	MI01187	WP219
1055	Copper	µg/L	Acceptable	175.	183	164 - 203	EPA 200.8.5.4 1994	184	9.44	MI01187	WP219
1070	Iron	µg/L	Acceptable	1660.	1610	1420 - 1810	EPA 200.8.5.4 1994	1630	79.2	MI01187	WP219
1075	Lead	µg/L	Acceptable	675.	714	624 - 801	EPA 200.8.5.4 1994	715	32.9	MI01187	WP219
1090	Manganese	µg/L	Acceptable	1180.	1110	998 - 1230	EPA 200.8.5.4 1994	1130	47.3	MI01187	WP219
1100	Molybdenum	µg/L	Acceptable	98.3	97.1	77.2 - 116	EPA 200.8.5.4 1994	96.6	4.28	MI01187	WP219
1105	Nickel	µg/L	Acceptable	1080.	1050	945 - 1170	EPA 200.8.5.4 1994	1040	48.5	MI01187	WP219
1140	Selenium	µg/L	Acceptable	822.	846	672 - 979	EPA 200.8.5.4 1994	841	38.9	MI01187	WP219
1150	Silver	µg/L	Acceptable	94.8	93.4	79.6 - 107	EPA 200.8.5.4 1994	93.3	4.72	MI01187	WP219
1160	Strontium	µg/L	Acceptable	247.	250	218 - 283	EPA 200.8.5.4 1994	252	10.6	MI01187	WP219
1165	Thallium	µg/L	Acceptable	185.	197	140 - 250	EPA 200.8.5.4 1994	195	12.7	MI01187	WP219
1185	Vanadium	µg/L	Acceptable	1690.	1670	1460 - 1860	EPA 200.8.5.4 1994	1650	80.2	MI01187	WP219
WP Trace Metals (cat# 586) (Continued)											
1190	Zinc	µg/L	Acceptable	1570.	1530	1320 - 1750	EPA 200.8.5.4 1994	1520	74.2	MI01187	WP219

All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: M10020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Trace Metals (cat# 586)											
1000	Aluminum	µg/L	Acceptable	308.	316	232 - 403	EPA 200.7 4.4 1994	331	22.8	MI01187	WP219
1015	Barium	µg/L	Acceptable	1780.	1900	1650 - 2140	EPA 200.7 4.4 1994	1890	77.7	MI01187	WP219
1020	Beryllium	µg/L	Acceptable	40.7	40.3	32.9 - 45.7	EPA 200.7 4.4 1994	39.7	2.26	MI01187	WP219
1030	Cadmium	µg/L	Acceptable	565.	608	519 - 690	EPA 200.7 4.4 1994	597	24.5	MI01187	WP219
1040	Chromium	µg/L	Acceptable	350.	358	311 - 406	EPA 200.7 4.4 1994	358	15.6	MI01187	WP219
1050	Cobalt	µg/L	Acceptable	438.	424	372 - 476	EPA 200.7 4.4 1994	438	17.7	MI01187	WP219
1055	Copper	µg/L	Acceptable	188.	183	164 - 203	EPA 200.7 4.4 1994	184	9.44	MI01187	WP219
1070	Iron	µg/L	Acceptable	1560.	1610	1420 - 1810	EPA 200.7 4.4 1994	1630	79.2	MI01187	WP219
1090	Manganese	µg/L	Acceptable	1050.	1110	998 - 1230	EPA 200.7 4.4 1994	1130	47.3	MI01187	WP219
1100	Molybdenum	µg/L	Acceptable	95.1	97.1	77.2 - 116	EPA 200.7 4.4 1994	96.6	4.28	MI01187	WP219
1105	Nickel	µg/L	Acceptable	985.	1050	945 - 1170	EPA 200.7 4.4 1994	1040	48.5	MI01187	WP219
1150	Silver	µg/L	Acceptable	94.4	93.4	79.6 - 107	EPA 200.7 4.4 1994	93.3	4.72	MI01187	WP219
1165	Thallium	µg/L	Acceptable	227.	197	140 - 250	EPA 200.7 4.4 1994	195	12.7	MI01187	WP219
1185	Vanadium	µg/L	Acceptable	1600.	1670	1460 - 1860	EPA 200.7 4.4 1994	1650	80.2	MI01187	WP219
1190	Zinc	µg/L	Acceptable	1460.	1530	1320 - 1750	EPA 200.7 4.4 1994	1520	74.2	MI01187	WP219
WP Mercury (cat# 574)											
1095	Mercury	µg/L	Acceptable	11.9	10.8	6.65 - 14.6	EPA 245.1 3 1994	10.6	1.12	MI01187	WP219
WP Low-Level Mercury (cat# 896)											
1095	Low Level Mercury	ng/L	Acceptable	22.8	21.5	11.0 - 32.0	EPA 1631E 2002	21.9	2.70	MI01187	WP219
DMRQA Low-Level Mercury (cat# 896)											
1095	Low Level Mercury	ng/L	Acceptable	88.57	74.3	56.5 - 91.2		79.2	6.24	MI00041	DMRQA33
WP Hexavalent Chromium (cat# 898)											
1045	Hexavalent Chromium	µg/L	Acceptable	170.	166	131 - 198	SM 3500-Cr B-2009 2009	166	10.1	MI01187	WP219

All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01



Page 7 of 12

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: M10020257
Permit Holder: Amy Vail

Lab Manager
Cadillac WWTP
1121 Plett Rd.
Cadillac, MI 49601
(231) 775-2368

ERA Customer Number: C454401
Report Issued: 07/29/13
Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Turbidity (cat# 893)											
2055	Turbidity	NTU	Acceptable	6.88	6.26	4.88 - 7.61	SM 2130 B-2001 2001	6.26	0.360	MI01187	WP219
WP Turbidity (cat# 893)											
2055	Turbidity	NTU	Acceptable	6.88	6.26	4.88 - 7.61	EPA 180.1.2-1993	6.26	0.360	MI01187	WP219
WP Total Cyanide (cat# 588)											
1645	Cyanide, total	mg/L	Acceptable	0.753	0.612	0.373 - 0.852	EPA 335.4 1993	0.600	0.0627	MI01187	WP219
WP Total Phenolics (4-AAP) (cat# 589)											
1905	Phenolics, total	mg/L	Acceptable	0.129	0.220	0.114 - 0.326	EPA 420.4.1-1993	0.239	0.0419	MI01187	WP219
WP Total Residual Chlorine (cat# 587)											
1940	Total Residual Chlorine	mg/L	Acceptable	2.00	1.80	1.29 - 2.22	SM 4500-Cl G-2000 2000	1.78	0.128	MI01187	WP219



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com





A Waters Company

Study: DMR-QA 33

NPDES Permit #: MI0020257

Laboratory Name: Cadillac WWTP

Microbiology Results





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: MI0020257

Permit Holder: Amy Vail

Lab Manager

Cadillac WWTP

1121 Plett Rd.

Cadillac, MI 49601

(231) 775-2368

ERA Customer Number: C454401

Report Issued: 07/29/13

Study Dates: 03/18/13 - 07/08/13

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
------------------	---------	-------	------------------------	----------------	----------------	-------------------	--------------------	------------	--------------------------	----------------	-------

DMRQA WasteWatr™ Coliform MicrobE™ (cat# 576)

2530	Fecal Coliforms (MF)	CFU/100mL	Acceptable	420	726	67.0 - 1450	SM 9222 D (m-FC)-1997 1997	313	231	MI00082	DMRQA33
------	----------------------	-----------	------------	-----	-----	-------------	-------------------------------	-----	-----	---------	---------

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.

WP WasteWatr™ Coliform MicrobE™ (cat# 576)

2500	Total Coliforms (MF)	CFU/100mL	Acceptable	1800	1490	290 - 4090	EPA 1604 MIAgar 2002	1090	634	MI01187	WP219
2525	E.coli (MF)	CFU/100mL	Acceptable	1060	1490	95.0 - 4380	EPA 1604 MIAgar 2002	646	713	MI01187	WP219

Per Section 6.4.3 a of the 2009 TNI Standard, "The assigned values for quantitative microbiology analytes shall be equal to the mean of the assigned value verification and/or homogeneity testing per Sections 7.1 and 7.2". The final acceptance limits are derived from the calculated study mean and study standard deviation from laboratory-reported results. Disagreement between the assigned values and study means/acceptance limits are due to the inherent variability of microbiology methods and differences in the methods used by ERA and participant laboratories. For quantitative microbiology analytes, the assigned value is not used in the evaluation of laboratories.



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com



Page 10 of 12



A Waters Company

Study: DMR-QA 33

NPDES Permit #: MI0020257

Laboratory Name: Cadillac WWTP

WET Results





DMR-QA 33 Final Report

A Waters Company

NPDES Permit #: MI0020257

Permit Holder: Amy Vail

Lab Manager

Cadillac WWTP

1121 Pielt Rd.

Cadillac, MI 49601

(231) 775-2368

ERA Customer Number: C454401

Report Issued: 07/29/13

Study Dates: 03/18/13 - 07/08/13

Analyte Code	Test End Point	Performance Evaluation	Reported Value %	Assigned Value %	Acceptance Limits %	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA Fathead minnow (Test Code 13) (cat# WET002) 48Hr., Acute, Non-Renewal, 25° C, MHSF Potassium chloride										
0754	LC50	Acceptable	72.9	57.0	29.5 - 84.5	EPA 2000.0 2002	57.0	13.8	MI01187	DMRQA33
DMRQA Fathead minnow (Test Code 15) (cat# WET004) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0808	IC25 (ON) Growth	Acceptable	57.0	50.1	26.9 - 73.4	EPA 1000.0 2002	50.1	11.6	MI01187	DMRQA33
0810	NOEC (ON) Growth	Acceptable	50.0	50.0	12.5 - 100	EPA 1000.0 2002			MI01187	DMRQA33
0756	NOEC Survival	Acceptable	50.0	50.0	25.0 - 100	EPA 1000.0 2002			MI01187	DMRQA33
DMRQA Ceriodaphnia dubia (Test Code 19) (cat# WET008) 48Hr., Acute Renewal, 25° C, MHSF Potassium chloride										
0764	LC50	Acceptable	52.3	49.4	24.7 - 74.1	EPA 2002.0 2002	49.4	12.3	MI01187	DMRQA33
DMRQA Ceriodaphnia dubia (Test Code 21) (cat# WET010) 7-day Short term Chronic, Daily Renewal, MHSF Potassium chloride										
0767	IC25 Reproduction	Acceptable	37.9	39.1	8.91 - 69.3	EPA 1002.0 2002	39.1	15.1	MI01187	DMRQA33
0768	NOEC Reproduction	Acceptable	25.0	25.0	12.5 - 50.0	EPA 1002.0 2002			MI01187	DMRQA33
0766	NOEC Survival	Acceptable	50.0	50.0	25.0 - 100	EPA 1002.0 2002			MI01187	DMRQA33
DMRQA Daphnia magna (Test Code 32) (cat# WET012) 48Hr., Acute, Non-Renewal, 25° C, MHSF Potassium chloride										
0788	LC50	Acceptable	72.3	61.6	36.8 - 86.4	EPA 2021.0 2002	61.6	12.4	MI01187	DMRQA33



All analytes are included in ERA's A2LA accreditation. Lab Code: 1539-01

16341 Table Mountain Pkwy • Golden, CO 80403 • 800.372.0122 • 303.431.8454 • fax 303.421.0159 • www.eraqc.com

